

Layla A. Schuster

laylaaschuster@gmail.com · laylaaschuster.github.io · [LinkedIn](#) · [ORCID](#)

Education

Doctor of Philosophy, Synthetic Biology

2022

University of Florida, Gainesville, FL

Department of Microbiology and Cell Science

Dissertation: "Promoter Toolkits to Study and Control Gene Expression Across the Proteobacteria"

Advisors: Christopher R. Reisch and Raquel Dias

Bachelor of Science in Microbiology and Cell Science

2013

University of Florida, Gainesville, FL

Current Position

Data Management Analyst III

Present

Department of Microbiology and Cell Science, University of Florida

Research Interests

Machine learning and explainable AI applied to biological sequence analysis; predicting and understanding promoter function across phylogenetically diverse bacterial species; broad host range genetic tools for synthetic biology; interpretable models that return biological insight; full-stack development of biological databases and research tools.

Publications

Underline indicates presenting/corresponding author. * indicates equal contribution.

Under Review

1. Schuster, L.A., Silva, J.C.F., Cheng, X., Liu, B., Dias, R., Zhao, M. Recombination and chromatin environment are central determinants of subgenome evolution in polyploids. *Science Advances* (under review).

Peer-Reviewed Articles

2. Silva, J.C.F., Richardson, M., Cediél-Becerra, J.D.D., Schuster, L., Dias, R. (2026). SpheronizaTor: Spherical voxelization for interpretable protein microenvironment modeling. *Computational and Structural Biotechnology Journal*, 35:0076. doi:10.34133/csbj.0076
3. Silva, J.C.F., Schuster, L., Sexson, N., Erdem, M., Hulke, R., Kirst, M., Resende, M.F.R., Dias, R. (2025). InteracTor: Feature engineering and explainable AI for profiling protein structure-interaction-function relationships. *PLOS Computational Biology*, 21(10), e1013038.

4. Franco Meléndez, K., **Schuster, L.**, Donahey, M.C., Kairalla, E., Jansen, M.A., Reisch, C., Rivers, A.R. (2024). MicroMPN: Methods and software for high-throughput screening of microbe suppression in mixed populations. *Microbiology Spectrum*, 12(3), e03578-23.
5. **Schuster, L.A.**, Reisch, C.R. (2022). Plasmids for controlled and tunable high-level expression in *E. coli*. *Applied and Environmental Microbiology*, 88(22), e00939-22.
6. **Schuster, L.A.**, Reisch, C.R. (2021). A plasmid toolbox for controlled gene expression across the Proteobacteria. *Nucleic Acids Research*, 49(12), 7189-7202.
7. Wang, G.P., Schnell, G.L., Kort, J.J., Sidhu, G.S., **Schuster, L.**, Tripathi, R.L., et al. (2021). Linkage of resistance-associated substitutions in GT1 sofosbuvir + NS5A inhibitor failures treated with glecaprevir/pibrentasvir. *Journal of Hepatology*, 75(4), 820-828.
8. Sidhu, G., **Schuster, L.**, Liu, L., Tamashiro, R., Li, E., Langae, T., Wagner, R., Wang, G.P. (2020). A single variant sequencing method for sensitive and quantitative detection of HIV-1 minority variants. *Scientific Reports*, 10(1), 8185.
9. Brown, A.N., Liu, L., Rodriguez, J.L., Zhao, L., **Schuster, L.**, Li, E., Wang, G.P., Neely, M.N., Yamada, W., Drusano, G.L. (2017). Sofosbuvir (SOF) suppresses ledipasvir (LDV)-resistant mutants during SOF/LDV combination therapy against genotype 1b hepatitis C virus (HCV). *Scientific Reports*, 7(1), 14421.

Preprints

10. **Schuster, L.**, Mejia, C., Rodriguez, L.T., Kairalla, E., Reisch, C.R., Chevrette, M.G., Dias, R. (2023). Characterizing constitutive promoters across the Proteobacteria. *bioRxiv*, 2023-11. [Manuscript in preparation]
11. Kalkreuter, E., Kautsar, S.A., Yang, D., Bader, C.D., Tejjaro, C.N., ..., **Schuster, L.**, et al. (2024). The Natural Products Discovery Center: Release of the first 8490 sequenced strains for exploring Actinobacteria biosynthetic diversity. *bioRxiv*, 2023-12.

Thesis

12. **Schuster, L.A.** (2022). Promoter toolkits to study and control gene expression across the Proteobacteria. PhD Dissertation, University of Florida.

Research Projects & Software

Machine Learning-Powered Promoter Design— Co-Investigator In Progress

Trained XGBoost classifiers on promoter sequence data across 15 Proteobacterial species (AUC >0.93). Applied SHAP-based explainability to identify species-specific sequence determinants of promoter activity. Manuscript in preparation.

Bark & Ambrosia Beetle Gallery — Co-Lead Full Stack Developer 2025–Present

Designed and built the frontend, backend, and database architecture for the world's largest curated database of bark and ambrosia beetle images and specimen data. Maintained by the UF Forest Entomology Lab. barkandambrosiagallery.org

Subgenome Evolution in Polyploids — First Author 2024–Present

Compiled genomic and epigenomic features for duplicated genes in maize and *Brassica rapa*. Trained XGBoost classifiers and applied SHAP interaction network analysis to identify recombination as a central meta-feature of subgenome dominance. Under review at Science Advances.

Technical Skills

Machine Learning & XAI: XGBoost, scikit-learn, SHAP, feature engineering, mutual information-based feature selection, model interpretation

Programming: Python (pandas, NumPy, scikit-learn, Matplotlib, seaborn), R, JavaScript, HTML/CSS, SQL

Bioinformatics: Sequence feature encoding, promoter analysis, genomic/epigenomic data processing

Wet Lab: Cloning and plasmid construction, PCR, library construction and screening, fluorescence assays, bacterial transformation, multi-species microbiology

Tools & Platforms: Git/GitHub, HiPerGator HPC, Linux/bash, Optuna hyperparameter optimization

Presentations

SynBio Young Speaker Series (SynBYSS) 2026

“Machine Learning-Powered Tools to Elucidate Species-Specific Promoter Design Rules Across the Proteobacteria”. Barcelona, Spain.

Plant & Animal Genome Conference 2026 (PAG 33) 2026

“Explainable AI Reveals Recombination and Chromatin Environment as Central Determinants of Subgenome Evolution in Polyploids”. San Diego, CA, USA.

International Society for Computational Biology (ISMB) 2024

“Machine Learning-Powered Tools to Elucidate Species-Specific Promoter Design Rules Across the Proteobacteria”. Montreal, Quebec, Canada.

Polyploidy Across the Tree of Life 2023

“Biased Fractionation in Promoter Regions of Maize”. Palm Coast, FL, USA.

References

Available upon request.